



**JAIN**  
DEEMED-TO-BE UNIVERSITY

FACULTY OF  
ENGINEERING  
AND TECHNOLOGY

**FACULTY OF ENGINEERING AND TECHNOLOGY**

**LAB EXERCISE - 5**

|                                  |                        |
|----------------------------------|------------------------|
| Program: CSE * C/GENERAL - E     | Semester: IV           |
| Course Name: WEB<br>TECHNOLOGIES | Course Code : 23CSE404 |
| Submission Date : 26/03/2026     | Year : II              |
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**5.) Add scientific calculator as a menu in your Exercise-1 using java script.**

**calculator.html:**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Modern Scientific Calc</title>
  <style>
    body {
      font-family: 'Segoe UI', sans-serif;
      background: #1e1e2f;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }
    .calc-container {
      background: #2d2d44;
      padding: 25px;
      border-radius: 20px;
      box-shadow: 0 10px 30px rgba(0,0,0,0.5);
      width: 350px;
    }
    #screen {
      width: 100%;
      height: 60px;
      background: #161625;
      border: none;
      color: #00ffcc;
      font-size: 24px;
      text-align: right;
      padding: 10px;
      box-sizing: border-box;
      border-radius: 10px;
      margin-bottom: 20px;
    }
    .grid {
      display: grid;
      grid-template-columns: repeat(4, 1fr);
      gap: 10px;
    }
  </style>
</head>
<body>
  <div class="calc-container">
    <div id="screen">
      <div class="grid">
        <div>+</div>
        <div>-</div>
        <div>*</div>
        <div>/</div>
        <div>=</div>
        <div>C</div>
        <div>1</div>
        <div>2</div>
        <div>3</div>
        <div>4</div>
        <div>5</div>
        <div>6</div>
        <div>7</div>
        <div>8</div>
        <div>9</div>
        <div>0</div>
        <div>.</div>
        <div>(</div>
        <div>)</div>
        <div>^</div>
        <div>√</div>
        <div>sin</div>
        <div>cos</div>
        <div>tan</div>
        <div>log</div>
        <div>ln</div>
        <div>exp</div>
        <div>1/x</div>
        <div>1/y</div>
        <div>1/z</div>
        <div>1/w</div>
        <div>1/v</div>
        <div>1/u</div>
        <div>1/t</div>
        <div>1/s</div>
        <div>1/r</div>
        <div>1/q</div>
        <div>1/p</div>
        <div>1/o</div>
        <div>1/n</div>
        <div>1/m</div>
        <div>1/l</div>
        <div>1/k</div>
        <div>1/j</div>
        <div>1/i</div>
        <div>1/h</div>
        <div>1/g</div>
        <div>1/f</div>
        <div>1/e</div>
        <div>1/d</div>
        <div>1/c</div>
        <div>1/b</div>
        <div>1/a</div>
        <div>1</div>
      </div>
    </div>
  </div>
</body>
</html>
```

```

    button {
      height: 50px;
      border-radius: 8px;
      border: none;
      background: #3d3d5c;
      color: white;
      font-size: 16px;
      cursor: pointer;
      transition: 0.2s;
    }
    button:hover {
      background: #50507a;
    }
    .btn-op { background: #ff9f0a; }
    .btn-sci { background: #5e5ce6; }
    .btn-equal {
      grid-column: span 2;
      background: #34c759;
    }
  </style>
</head>
<body>

<div class="calc-container">
  <input type="text" id="screen" readonly>
  <div class="grid">
    <button class="btn-sci" onclick="push('Math.sin()')>sin</button>
    <button class="btn-sci" onclick="push('Math.cos()')>cos</button>
    <button class="btn-sci" onclick="push('Math.sqrt()')>√</button>
    <button class="btn-sci" onclick="push('**')>^</button>

    <button onclick="push('7')>7</button>
    <button onclick="push('8')>8</button>
    <button onclick="push('9')>9</button>
    <button class="btn-op" onclick="clr()">AC</button>

    <button onclick="push('4')>4</button>
    <button onclick="push('5')>5</button>
    <button onclick="push('6')>6</button>
    <button class="btn-op" onclick="push('/')>÷</button>

```

```
<button onclick="push('1')">1</button>
<button onclick="push('2')">2</button>
<button onclick="push('3')">3</button>
<button class="btn-op" onclick="push('*')">×</button>

<button onclick="push('0')">0</button>
<button onclick="push('.')">.</button>
<button class="btn-op" onclick="push('-)">-</button>
<button class="btn-op" onclick="push('+)">+</button>

<button onclick="push('(')">(</button>
<button onclick="push(')'">)</button>
<button class="btn-equal" onclick="solve()">=</button>
</div>
</div>

<script>
  const screen = document.getElementById('screen');

  function push(val) {
    screen.value += val;
  }

  function clr() {
    screen.value = "";
  }

  function solve() {
    try {
      // Using Function constructor instead of eval for better security practice
      screen.value = new Function('return ' + screen.value)();
    } catch (err) {
      screen.value = "Error";
    }
  }
</script>

</body>
</html>
```

**Output:**



